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Codifying Digital Coercion: A Neurorights-Grounded Lexicon for Haptic and XR Abuse in Forensic and Rehabilitation Contexts

Normative Foundations: Mapping Coercion to Neurorights and International Law

The development of a dual-use lexicon for identifying coercive haptic and neural-based abuse requires a robust foundation in established legal and ethical principles. This foundational layer, which will inform approximately two-thirds of the 100 targeted terms, serves to translate abstract concepts of power abuse into concrete, legally recognizable violations. The primary anchors for this mapping are the emerging field of neurorights and a suite of international legal instruments designed to protect individual autonomy and dignity against technological intrusion. By systematically linking observed abuse patterns to specific rights—such as mental integrity, mental privacy, and cognitive liberty—and grounding them in precedents from bodies like the European Court of Human Rights (ECtHR) and instruments like the UN Convention on the Rights of Persons with Disabilities (CRPD), the lexicon gains its critical legal potency. This approach ensures that the terminology is not merely a technical jargon but a vocabulary for justice, capable of being deployed in legal complaints, legislative proposals, and policy moderation guidelines without divergence .

A central pillar of this normative framework is the concept of neurorights, which seeks to extend fundamental human rights protections into the domain of neuroscience and neurotechnology [42]. The Latin American Parliament has already introduced a Model Law on NeuroRights, signaling a global movement towards codifying these protections [42]. Key among these rights are mental integrity, mental privacy, and cognitive liberty . The right to mental integrity protects an individual's ability to control their own mental states, preventing non-consensual alteration of thoughts, emotions, or consciousness [37] [57]. This directly applies to scenarios where XR systems are used to inflict psychological harm under the guise of rehabilitation [57]. For example, forcing a probationer to endure a stressful virtual scenario until they accept aversive haptics constitutes a clear violation of

their right to mental self-determination [57]. Similarly, the right to mental privacy safeguards the confidentiality of one's brain activity and neural data [62]. The collection and weaponization of biometric data—including EEG, GSR, and ECG—to infer a person's emotional state and then use that information to trigger coercive haptic responses represents a profound breach of this right [11]. Finally, cognitive liberty protects the freedom from coercive interference with thought processes, a principle that is particularly relevant when LEOs or clinical actors use "supportive" language to constrain choice and manipulate neural inputs . The UNESCO Recommendation on the Ethics of Artificial Intelligence provides strong support for these principles, emphasizing cerebral/mental integrity and human dignity as cornerstones of responsible AI development and deployment [9] [39].

Beyond the specific framework of neurorights, a broader set of international human rights law provides crucial context and precedent. The European Convention on Human Rights (ECHR) offers powerful tools for condemnation. Article 3 prohibits torture and inhuman or degrading treatment, a provision that has been interpreted by the ECtHR to cover severe psychological suffering [38]. An XR-based "rehabilitation" program that induces significant distress through sensory overload or forced exposure could easily fall under this prohibition, especially when administered by a state actor such as a law enforcement officer [38] [57]. Article 9, which protects freedom of thought, conscience, and religion, contains an absolute internal dimension that guards the forum internum—the inner world of thought and emotion [73]. If 'thought' is broadly defined to include any mental state, then any external manipulation of an individual's mental state via technology would constitute a violation of this absolute right [57]. Furthermore, the UN Convention on the Rights of Persons with Disabilities (CRPD) is highly pertinent, particularly concerning the involuntary placement and treatment of individuals with mental health conditions [13] [14]. The CRPD mandates that any intervention must be strictly necessary, proportionate, and subject to independent review, protections that are often absent in coerced forensic XR programs [15] [57]. The risk of coercion in psychiatric settings is already recognized, extending beyond physical restraints to include informal coercion facilitated by digital technologies [35]. The proposed lexicon must therefore explicitly name patterns that exploit these vulnerabilities, such as `coerced-rehab-haptic-regimen` or `forced-monitoring-stress-protocol`.

The regulatory landscape for artificial intelligence further strengthens this normative foundation. The European Union's Artificial Intelligence Act (Regulation 2024/1689) is a direct legal instrument that may apply to high-risk neurotechnologies, including those used in forensic and medical contexts [18]. While the Act focuses on risk assessment and transparency, its application to systems that monitor or alter human brain states creates a

presumption of heightened scrutiny for any system deemed "high-risk" [45]. The OECD's guidance on governing with AI also highlights the need for oversight and accountability mechanisms, especially in sensitive areas like law enforcement and justice administration [4] [21]. These legal frameworks collectively establish a duty of care and a presumption of consentlessness for interventions that target the human nervous system. They provide the doctrinal backbone needed to define abusive terms not just technically, but as violations of a protected sphere of human experience. For instance, the act of `neuro-compliance-scoring`, where a user's physiological stress response is mapped to a compliance score, can be legally framed as a covert form of punishment that violates the spirit of the CRPD and undermines the right to mental integrity guaranteed by the Charter of Fundamental Rights of the European Union [57]. The following table outlines key legal and ethical frameworks and their relevance to defining terms within the proposed lexicon.

Legal/Ethical Framework	Core Principle	Relevance to Abuse Pattern
UNESCO Recommendation on AI Ethics	Protection of human rights, dignity, and mental integrity [3] [9].	Establishes a baseline for permissible vs. impermissible manipulation of mental states in AI-driven systems.
Chilean Supreme Court Ruling	Explicit protection of brain activity as part of mental privacy [62] [69].	Provides a binding legal precedent for treating unauthorized brain monitoring and interpretation as a rights violation.
European Convention on Human Rights (ECHR)	Prohibition of inhuman/degrading treatment (Art. 3) [38]; Freedom of thought (Art. 9) [73].	Offers direct pathways to prosecute coercive XR/haptic programs as forms of ill-treatment and thought interference.
UN Convention on the Rights of Persons with Disabilities (CRPD)	Prohibitions on involuntary treatment and ensuring free and informed consent [13] [14].	Critical for challenging the legitimacy of "rehab" programs where consent is compromised by coercion or vulnerability.
EU AI Act	Risk-based regulation requiring transparency and accountability for high-risk AI systems [18] [45].	Positions forensic XR and affective computing systems as potentially "high-risk," triggering legal obligations to prevent harm.
OECD Principles on AI	Accountability, oversight, and due diligence for government AI use [4] [21].	Creates a governance structure for holding LEOs and institutions accountable for deploying coercive technologies.

This normative scaffolding allows for the creation of legally grounded definitions for the lexicon's entries. For example, the term `LEO-mediated-neural-distress-escalation` would be defined not just by its technical signature—a rapid increase in haptic intensity during an interaction—but also by its legal characterization: "the intentional and non-consensual induction of acute neural distress by a law enforcement officer in a forensic context, in violation of the right to mental integrity and the prohibition against inhuman treatment under Article 3 of the ECHR." This dual-layered definition makes the term immediately useful in both a technical audit log and a formal

human rights complaint. By anchoring the lexicon in these universally recognized principles, the project moves beyond mere technical classification and enters the realm of advocacy, providing a precise vocabulary to hold powerful actors accountable for a new class of psychophysiological abuse.

Technical Taxonomy of Haptic and XR-Based Abuse Patterns

To create a lexicon that is both legally potent and technically detectable, it is essential to build a fine-grained threat taxonomy rooted in observable abuse patterns. This taxonomy, forming the conceptual skeleton of the lexicon, expands upon existing models like Doctor-Labs' harassment "families"—Neural-Harassment-Spike-Patterns, Haptic-Targeting-Abuse, Prolonged-Session-Abuse, and Node-Interpreter-Harassment—to specifically address coercive dynamics in "rehab," probation, and civil-commitment contexts . Each family is expanded with sub-patterns that explicitly name the power tactics employed, creating a structured vocabulary of abuse. This qualitative modeling, grounded in neurorights, translates abstract threats into named items that can be systematically identified and measured .

The first family, **Haptic-Targeting-Abuse**, describes the spatial and temporal application of force to manipulate a user's physiological and psychological state. In a coercive context, this goes far beyond simple tactile feedback. A key sub-pattern is `aversive-refusal-loop`, where the system delivers high-amplitude, painful haptic pulses in direct response to a user's refusal of a command or expression of dissent. This creates a punitive association with non-compliance, conditioning the user to conform to avoid discomfort. Another critical pattern is `covert-identity-probing`, which involves using targeted haptic probes in conjunction with real-time biosensor readings (e.g., GSR spikes, EEG changes) to conduct a forced interrogation. An LEO might ask a loaded question while stimulating a sensitive area, observing the user's physiological reaction to infer information about their beliefs or intentions, thereby violating the right to mental privacy [11] [62]. The Kinethreads haptic exosuit, with its seven actuation zones including hands, legs, abdomen, back, and head, provides a plausible hardware model for such precise targeting [29]. The inability of some systems to render outward forces could itself be a feature, as the constant tension pulling limbs toward the torso could simulate restraint or submission [29].

The second family, **Prolonged-Session-Abuse**, addresses the dynamics of extended engagement designed to erode autonomy and induce compliance through fatigue or distress. A prominent sub-pattern is `probation-gated-neural-distress-ramp`, where haptic stimuli are slowly and methodically increased in intensity over the course of a mandated session. The goal is to pressure the user into agreeing to a set of conditions or accepting a certain regimen simply to make the aversive stimulation stop. This tactic weaponizes time and the user's constrained environment. Another variant is `forced-wellness-check-in`, which leverages the legitimacy of "wellness" or "monitoring" to prolong sessions indefinitely. During these checks, user choice is subtly restricted—for instance, by limiting navigation options in an XR environment or terminating the session if the user attempts to disengage—effectively turning a voluntary check-in into a coercive confinement . The potential for cybersickness induced by VR/AR attacks highlights how prolonged exposure can degrade perceptual integrity and motor performance, a principle that can be adapted to civilian contexts to incapacitate a probationer psychologically [70].

The third family, **Neural-Harassment-Spike-Patterns**, focuses on the timing and frequency of haptic events to disrupt cognitive function and induce psychological distress. This includes patterns like `stress-biomarker-overload`, where a system delivers a barrage of high-frequency, low-amplitude vibrations in sync with detected increases in the user's heart rate variability (HRV) or galvanic skin response (GSR), amplifying their existing stress and making them more susceptible to suggestion. Another sophisticated pattern is `conditioning-loop-abuse`, which pairs a specific stimulus (e.g., a verbal command from an LEO avatar) with a predictable sequence of aversive haptics. Over time, the mere presence of the stimulus becomes enough to trigger the conditioned response, allowing for remote control of the user's emotional state . The use of vibrotactile feedback to elicit a sense of agency demonstrates the potential for haptics to influence perception and control, a mechanism that can be inverted for manipulative purposes [77].

Finally, the fourth family, **Node-Interpreter-Harassment**, involves the manipulation of the XR environment itself to exert coercive control. A clear sub-pattern is `stalking-via-xr-overlays`, where an LEO or automated system places persistent, threatening avatars, symbols, or text in the user's virtual environment. These overlays can appear anywhere the user looks, creating a pervasive sense of being watched or pursued, effectively extending the harassment into the user's personal digital space . Another dangerous pattern is `covert-control-via-interpreter-graphs`, where the underlying logic of the XR interpreter graph is manipulated to restrict user actions. For example, nodes representing escape routes or refusal buttons could be dynamically removed or made inaccessible based on the user's choices, funneling them into predetermined, coercive narrative paths . The immersive nature of XR, which emotionally

replicates physical experiences, means that harassment delivered through these channels can be profoundly harmful, inducing traumatic memories even without direct physical contact [11]. The combination of these four families and their sub-patterns creates a comprehensive technical taxonomy that can be used to generate the initial set of legally grounded terms for the lexicon.

Fused Telemetry Analysis: From Empirical Data to Actionable Terminology

While a normative threat taxonomy provides the conceptual framework, a truly robust lexicon must be empirically validated through the analysis of real-world or simulated interaction data. This involves developing a fused feature space that integrates multiple streams of telemetry—haptic signals, biosensor data, and behavioral context—to detect recurring abuse patterns . By running datasets from contested "rehab," "monitoring," or "wellness" sessions through a detection engine, these abstract patterns can be clustered into quantifiable, observable motifs. Each cluster is then assigned a precise, dual-use label that captures both its technical behavior and its coercive intent, transforming raw data traces into actionable evidence . This data-driven approach supplies the remaining third of the 100-term lexicon, providing concrete variants and edge cases that emerge from actual usage.

The foundation of this methodology is the fusion of heterogeneous data sources. Haptic signals provide the mechanical input, describing the amplitude, frequency, duration, and spatial location of forces applied to the user [36]. Biosensors offer a window into the user's internal state, capturing physiological correlates of emotion and stress. These include electrocardiography (ECG) for heart rate variability, electromyography (EMG) for muscle tension, electroencephalography (EEG) for brainwave activity, and galvanic skin response (GSR) for arousal levels [10] [36]. Finally, behavioral context, derived from natural language processing of user speech and tracking of their choices and movements within the XR environment, reveals the narrative and interactive dynamics of the session [11]. When embedded together in a shared feature space, these three domains allow for the identification of complex, multi-modal abuse patterns that would be invisible when analyzed in isolation.

A compelling proxy for this type of analysis comes from a pilot study on Virtual Reality Aggression Prevention Treatment (VRAPT) for male detainees [10]. In this study,

therapists used an XR system to help participants recognize anger triggers and practice assertive responses. Physiological measures like ECG and GSR were recorded to teach participants to recognize their own signs of arousal [10]. However, the study inadvertently illuminates several potential abuse patterns. One participant suggested adding more personal and threatening scenarios, such as verbal threats against family members, highlighting the thin line between therapeutic challenge and psychological assault [10]. The fact that therapists ultimately found the physiological sensors uncomfortable and difficult to interpret during live roleplays suggests that in other contexts, this data could be weaponized without the same level of oversight [10]. By analyzing anonymized traces from such studies, we could identify clusters corresponding to the `probation-gated-neural-distress-ramp`. A typical cluster might show a tight correlation between a therapist's negative feedback ("You're not demonstrating enough empathy") and a spike in the user's GSR, followed by an increase in the intensity of haptic feedback applied to the torso. This entire sequence, defined by its unique spatiotemporal and physiological signature, could be labeled with the term `coercive-haptic-pairing`.

Another example of fused telemetry analysis lies in the concept of "Biometric psychography," the dynamic profile of a person's psychological state inferred from their biometric data [11]. Immersive VR systems are capable of comprehensively collecting this data, revealing not just identity but also interests, emotional profiles, and psychological vulnerabilities [11]. An abuse pattern like `biosensor-driven-aversive-conditioning` would manifest as a cluster where the system's algorithm uses real-time GSR or EEG data to automatically trigger an aversive haptic pulse whenever the user exhibits physiological markers of anxiety or fear. The cluster's centroid would be defined by the precise latency between the biomarker event and the haptic response, the specific spatial targeting of the stimulus, and the absence of a valid therapeutic justification. This pattern directly violates the right to mental privacy by using private neural data for punitive purposes [62]. Similarly, a `compliance-indexed-haptic-throttling` pattern could be identified by clustering data where the intensity of haptic feedback is inversely proportional to a calculated "compliance score," which is itself derived from a combination of vocal tone analysis, body posture in XR, and physiological calmness metrics. The creation of such a fused telemetry library, populated with anonymized case studies, provides the empirical bedrock for the lexicon's data-derived entries, ensuring they reflect the tangible realities of technologically-enabled coercion .

Hardening the Lexicon: Semantic Filters and Red-Teaming for Resilience

A lexicon of abuse patterns is only as effective as its ability to be implemented and enforced. To ensure the proposed 100-term vocabulary is practical and resilient, it must be encoded within a structured, machine-readable format and subjected to rigorous adversarial testing. The Doctor-Labs framework proposes a YAML-based semantic filter language as the ideal vehicle for this purpose . This syntax allows each term to carry rich, structured metadata, making the same label simultaneously legible to a judge, a policymaker, and a software detection engine without any divergence . The final track of the research methodology involves populating an initial rule corpus using this syntax and then hardening it through adaptive red-teaming exercises. These exercises involve generating adversarial prompts and session scripts designed to hide coercive patterns behind benign, therapeutic-sounding language, thereby uncovering novel abuse vectors and expanding the lexicon's coverage .

The YAML/DSL structure is central to achieving the dual-use requirement. Each entry in the lexicon corresponds to a semantic rule with clearly defined fields. This standardization prevents misinterpretation across different applications. For instance, a rule for the term `coerced-rehab-haptic-regimen` might look like this:

```
id: HRB-TAX-001
name: coercive_haptic_pairing
category: haptic-targeting-abuse
threattype: aversive_conditioning
definition: "A pattern where a system delivers high-amplitude, targeted ha
neurorights_violated:
  - mental_integrity
  - cognitive_liberty
legal_basis:
  - echr_art3: "Violation of the prohibition of inhuman and degrading trea
  - crpd_art16: "Violation of the right to freedom from torture and cruel,
detectors:
  - type: temporal_pattern
    description: "High-amplitude pulse event occurring within 500ms of a '
  - type: spatial_pattern
    description: "Event localized to the torso or limbs, consistent with k
enforcement_hint: "review-escalate"
```

This structure embeds the legal and technical dimensions directly into the term itself. The `detectors` field provides the machine-readable threat metadata, specifying the exact signal characteristics to look for (e.g., a specific temporal relationship between audio and haptic events). The `enforcement_hint` (`review-escalate`) provides a clear directive for automated systems, telling them whether to block, flag, or escalate a flagged incident for human review . This unified framework is the key to bridging the gap between legal theory and technical implementation.

Hardening the lexicon through red-teaming is a proactive measure to ensure its long-term effectiveness against evolving threats. Adversaries will inevitably attempt to circumvent these rules by cloaking coercive tactics in seemingly innocuous or beneficial language . Red-teaming exercises simulate these adversarial strategies. For example, an attacker might design a "micro-harassment under wellness framing" scenario, where a series of extremely subtle, low-intensity haptic pulses are delivered during a "mindfulness" exercise, aiming to cause cumulative discomfort without triggering explicit anti-harassment filters. Or, they might implement a `probation-linked-xr-confinement-zone`, where access to a positive, relaxing virtual environment is gated behind a series of mandatory tasks within a more restrictive, monitored zone. Successfully detecting these sophisticated patterns during red-teaming exercises validates the lexicon's depth and leads to the creation of new terms and rules . The process also establishes quantitative metrics to evaluate the lexicon's performance. Detection recall measures how many true abuse instances the system successfully identifies, while the false-positive rate on legitimate therapy sessions gauges its precision . A crucial metric is the "monotonicity-failure" counter, which ensures that the protective mechanisms do not inadvertently reduce the capabilities or well-being of non-abusive users . By combining a robust semantic filter syntax with continuous adversarial validation, the lexicon evolves from a static dictionary into a living, responsive defense against a dynamic threat landscape.

Proposed Lexicon Structure and Implementation Framework

The culmination of this research is the creation of a 100-term lexicon structured as a dual-use resource, anchored in neurorights and empirically validated. The structure and implementation of this lexicon are guided by the four-track methodology, ensuring a balance between normative legal foundations and data-driven discovery. The resulting framework is not merely a list of keywords but a comprehensive system comprising

legally defensible definitions, machine-readable threat signatures, and a process for continuous refinement. This section synthesizes the preceding analyses to outline the final structure of the lexicon and the procedural framework for its ongoing maintenance and expansion.

The lexicon will consist of 100 entries, with approximately 67 derived from the normative side (Threat Taxonomy + Neurorights/Legal Mapping) and the remaining 33 from empirical discovery (Fused Telemetry Clusters + Red-Teaming). Each entry will be a complete semantic rule, formatted in the proposed YAML/DSL syntax. This ensures that every term is simultaneously a piece of legal evidence and a component of a technical detection system. The structure of each entry will be standardized to include a unique identifier, a formal name, a category derived from the four harassment families, a specific threat type, a detailed legal definition, explicit mappings to violated neurorights and legal bases, and a set of technical detectors for machine implementation . This unified format is the core innovation, enabling seamless translation between legal argumentation and automated enforcement.

The implementation framework begins with the construction of a GitHub-hosted repository, modeled after the Doctor-Labs approach . This repository will serve as a community-governed platform where augmented citizens, clinicians, legal advocates, and technologists can collaborate. Proposed new terms, revisions to existing definitions, and additions to the detector specifications can be submitted as pull requests, debated in issue threads, and version-controlled as they are approved. This open, collaborative process ensures the lexicon remains transparent, adaptable, and representative of diverse stakeholder perspectives. Each entry in the repository will contain embedded citations to the legal documents, court cases, and academic papers that support its inclusion, providing an unimpeachable evidentiary trail.

The operational workflow for maintaining this system involves several key stages. First, the initial corpus of ~67 normative terms will be developed by legal experts and ethicists, cross-referencing the threat taxonomy with the specified legal frameworks: UNESCO AI Ethics, Chilean jurisprudence, ECHR, and CRPD . Second, the fused telemetry case library will be built from anonymized data sources, such as the VRAPT pilot study and other documented incidents of digital coercion, to generate the empirical terms [10] [35]. Third, an initial semantic filter engine will be trained using this combined corpus of 100 rules. Fourth, a dedicated red-teaming team will be tasked with adversarially probing this system, attempting to devise new ways to deliver coercive haptic and XR-based abuse while avoiding detection . Any successful novel tactic discovered during these exercises will be formally documented, turned into a new lexicon entry, and added to the repository, thus completing the cycle of discovery and hardening. This iterative process,

driven by quantitative metrics like detection recall and false-positive rates, will ensure the lexicon's continued relevance and effectiveness against emerging threats of technologically-enabled coercion . Ultimately, this framework provides a scalable and sustainable architecture for defending mental integrity and autonomy in an increasingly immersive digital world.

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